Section B – Employees

This section provides financial insights into our employee benefit arrangements. It should be read in conjunction with the compensation disclosures for key management personnel in *Note (G.5)* as well as *SAP's Compensation Report*.

(B.1) Employee Headcount

The following table provides an overview of employee headcount, broken down by function and by the regions EMEA (Europe, Middle East, and Africa), Americas (North America and Latin America), and APJ (Asia Pacific Japan).

Employee Headcount by Region and Function

Full-time equivalents	12/31/2019				12/31/2018				12/31/2017			
	EMEA	Ame-ricas	APJ	Total	EMEA	Ame-ricas	APJ	Total	EMEA	Ame-ricas	APJ	Total
Cloud and software	6,501	4,426	5,361	16,288	6,341	4,268	5,374	15,983	5,869	3,895	4,719	14,482
Services	8,250	6,018	5,971	20,239	8,120	5,736	5,620	19,476	7,536	4,878	4,965	17,379
Research and development	12,710	5,793	9,131	27,634	12,478	5,651	8,930	27,060	11,349	5,250	8,273	24,872
Sales and marketing	10,205	10,368	5,209	25,781	9,843	9,452	4,918	24,213	9,196	9,169	4,854	23,219
General and administration	3,161	2,123	1,246	6,530	2,906	1,970	1,147	6,024	2,676	1,781	1,047	5,504
Infrastructure	2,220	984	654	3,859	2,160	951	631	3,742	1,732	855	501	3,087
SAP Group (12/31)	**43,048**	**29,712**	**27,571**	**100,330**	**41,848**	**28,029**	**26,620**	**96,498**	**38,357**	**25,827**	**24,359**	**88,543**
Thereof acquisitions	338	1,638	137	2,113	657	952	434	2,043	149	133	7	289
SAP Group (months' end average)	**42,697**	**29,368**	**27,092**	**99,157**	**40,496**	**27,454**	**25,759**	**93,709**	**37,512**	**25,459**	**24,029**	**86,999**

(B.2) Employee Benefits Expenses

Components of Employee Benefits Expenses

€ millions	2019	2018	2017
Salaries	10,031	9,025	8,693
Social security expenses	1,477	1,339	1,281
Share-based payment expenses	1,835	830	1,120
Pension expenses	369	330	312
Employee-related restructuring expenses	1,111	19	180
Termination benefits outside of restructuring plans	47	52	57
Employee benefits expenses	**14,870**	**11,595**	**11,643**

(B.3) Share-Based Payments

Accounting for Share-Based Payments

Classification in the Income Statement

Share-based payments cover cash-settled and equity-settled awards issued to our employees. The respective expenses are recognized as employee benefits and classified in our Consolidated Income Statements according to the activities that the employees perform. Where we economically hedge our exposure to cash-settled awards, changes in the fair value of the respective hedging instruments are also recognized as employee benefits expenses in profit or loss. The fair values of hedging instruments are based on market data reflecting current market expectations. For more information about the equity price risk, see *Note (F.1)*.

Valuation, Judgment, and Sources of Estimation Uncertainty

We use certain assumptions in estimating the fair values for our share-based payments, including expected share price volatility and expected award life (which represents our estimate of the average remaining life until the awards are exercised or expire unexercised). In addition, the final payout for plans also depends on the achievement of performance indicators and on our share price on the respective exercise dates. Changes to these assumptions and outcomes that differ from these assumptions could require material adjustments to the carrying amount of the liabilities we have